Hybrid Sensor-Based Early Detection of Roadside Humans and Animals for Accident Prevention

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Abstract :

Road accidents caused by unexpected crossings of pedestrians and stray animals are a major global problem, with disproportionate impacts in semi-urban and rural regions. This paper presents a hybrid, low-cost Internet of Things (IoT) system designed for early detection of humans and animals near roadways and for rapid accident alerting. The system integrates an ESP32 microcontroller with a webcam for visual classification, PIR sensors for motion detection, ultrasonic sensors for proximity estimation, and an optional IMU/vibration sensor for impact confirmation. GPS and GSM modules provide geolocation and emergency communication respectively. Sensor fusion is performed on the ESP32 using a lightweight rule-based decision engine that balances detection sensitivity against false positives. When a high-risk event is identified, the system immediately issues an in-vehicle audio/visual warning, captures an image via the webcam (for visual evidence), and dispatches an SMS containing GPS coordinates to predefined emergency contacts. Prototype trials under varied lighting and environmental conditions show that the hybrid approach increases detection accuracy and reduces false alarms compared with single-sensor solutions; simulated results indicate detection accuracy >90% and false positive rates under 7% in tested scenarios. The paper details hardware selection, circuit overview, data fusion strategy, experimental procedure, and quantitative results, and discusses limitations and future enhancements including edge-AI models and radar integration.

Keywords :

IoT, hybrid sensors, webcam, ESP32, accident prevention, GPS, GSM, PIR, ultrasonic, sensor fusion.

I. Introduction :

Road accidents remain one of the leading causes of fatalities and injuries worldwide, posing a critical threat to both human lives and socio-economic development. According to the World Health Organization (WHO), more than 1.19 million people die annually due to road accidents, while millions more sustain serious injuries requiring long-term treatment. A considerable portion of these accidents involves pedestrians and strays animals, especially in semi-urban and rural regions where road infrastructure and safety enforcement are often inadequate. These incidents not only cause loss of life but also place a heavy financial burden on healthcare systems, insurance providers, and the families of victims. Traditional accident prevention systems, such as road signs, vehicle horns and basic lighting infrastructure, have proven insufficient in detecting and preventing Recent advances in Intelligent Transportation Systems (ITS) and

Internet of Things (IoT)-based safety solutions have shown promising outcomes in reducing accident risks. However, many of the existing solutions rely on single-sensor architectures, such as ultrasonic distance measurement or thermal imaging, which suffer from limitations under real-world environmental conditions. For example, ultrasonic sensors may misinterpret non-living roadside objects as obstacles, while thermal sensors may fail to accurately distinguish between humans and animals at long ranges.

In addition, most commercially available Advanced Driver Assistance Systems (ADAS) are designed for high-end vehicles, making them unaffordable for two-wheelers and budget four-wheelers, which are more vulnerable to roadside collisions. There is a clear need for a low-cost, hybrid, and scalable accident prevention system that can integrate into a wide range of vehicles, including motorcycles and low-budget cars commonly used in developing countries.

The proposed system addresses this challenge by combining multiple sensing modalities into a unified architecture. An ESP32 microcontroller serves as the central processing unit, interfacing with an ESP32-CAM webcam module for object recognition, a Passive Infrared (PIR) sensor for motion and heat-based detection, and an ultrasonic sensor for proximity measurement. A GPS module (NEO-6M) and a GSM module (SIM800L/SIM900A) provide real-time location tracking and emergency communication capabilities. Upon detecting a potential roadside human or animal, the system generates audio-visual alerts via buzzers and LEDs, giving the driver sufficient time to react. In the event of a collision or critical situation detected by the vibration sensor, the system automatically transmits an emergency SMS containing GPS coordinates to pre-registered contacts. This hybrid approach ensures greater detection accuracy and resilience compared to single-sensor solutions. By leveraging both vision-based and thermal/motion-based sensing, the system reduces false positives while maintaining low power consumption suitable for embedded automotive applications. Furthermore, the low hardware cost and modular design make it feasible for integration into existing vehicles without significant structural modification.

The rest of this paper is organized as follows: Section II presents the objectives of the proposed system. Section III provides a literature review of related research works published in 2023–2024. Section IV describes the methodology and detailed system design. Section V outlines the results and discussion of prototype testing.

II. Objectives :

The primary objective of this research is to design and implement a hybrid sensor-based intelligent system for the early detection of roadside humans and animals to prevent vehicular accidents. Unlike conventional single-sensor systems, the proposed framework combines multiple sensing technologies, ensuring better accuracy, reliability, and cost-effectiveness.

1. To enhance detection accuracy using a hybrid sensing approach

Traditional systems that rely on only one type of sensor often fail under specific environmental conditions. For instance, ultrasonic sensors can produce false triggers when detecting roadside poles or bushes, while vision-based systems may be less effective in low-light or foggy environments. By combining PIR motion sensing, ultrasonic ranging, and camera-based recognition, the proposed model aims to minimize such inaccuracies and improve the reliability of detection in real-world scenarios.

2. To integrate real-time accident alert mechanisms

In the event of an accident, timely communication with emergency contacts can be the difference between life and death. The objective here is to integrate GPS and GSM modules with the core system so that accurate location details can be sent via SMS to pre-registered numbers.This feature ensures that even in remote areas without immediate medical assistance, rescue operations can be mobilized quickly.

3. To provide driver alerts through audio-visual warning systems

Another important goal is to alert the driver before an accident occurs. This is achieved using buzzers and LED indicators, which notify the driver of the presence of humans, animals, or unexpected roadside movement. The alerts are designed to be non-intrusive but effective enough to catch the driver's attention instantly.

4. To design a cost-effective and energy-efficient solution

High-end Advanced Driver Assistance Systems (ADAS) are expensive and often limited to premium vehicles. The proposed system seeks to create a low-cost solution using readily available IoT components such as the ESP32 board, PIR sensors, ultrasonic sensors, and webcams. At the same time, the design emphasizes low power consumption, making it suitable for integration into motorcycles and budget-friendly vehicles without draining the battery.

5. To ensure system scalability and modularity

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system seeks to create a low-cost solution using readily available IoT components such as the ESP32 board, PIR sensors, ultrasonic sensors, and webcams. At the same time, the design emphasizes low power consumption, making it suitable for integration into motorcycles and budget-friendly vehicles without draining the battery.

6. To test and validate the system under real-world conditions

The project also aims to evaluate the proposed system in different road conditions such as highways, semi-urban streets, and rural areas where stray animals are common. The objective is to measure performance in terms of detection accuracy, false alarm rates, SMS alert latency, and overall driver response time. These results will help in validating the effectiveness and practicality of the system.

7. To contribute to road safety research and societal benefits

Beyond technical goals, this project aims to contribute to reducing accidents and saving lives, particularly in developing countries where affordable safety systems are critically needed. The broader objective is to demonstrate how IoT and hybrid sensor integration can play a transformative role in intelligent transportation systems and road safety management.

III. Literature Review :

[1] Road safety and accident prevention have been active areas of research for decades, with recent emphasis on IoT-enabled and AI-assisted solutions.

[2] Several studies highlight the growing role of multi-sensor integration to enhance accuracy in detecting humans, animals, and obstacles near roadways.

[3] In 2023, Singh et al. [1] proposed an IoT-based vehicle accident detection system using an Arduino controller, GSM, and GPS modules, demonstrating faster alert transmission compared to traditional reporting methods.

[4] Similarly, Rahman and Chowdhury [2] developed a smart accident detection system integrating vibration and GPS sensors, though the model was limited by false alerts in high-vibration environments.

[5] Recent advances in vision-based detection, such as those reported by Zhang et al. [3] in 2024, leverage low-power embedded cameras for real-time pedestrian recognition, showing that compact vision systems can be deployed even in cost-sensitive applications.

[6] However, vision-only models often face challenges in poor visibility conditions such as fog, rain, or nighttime driving.

[7] To address this, Khan et al. [4] combined ultrasonic and infrared sensors for obstacle detection, achieving improved accuracy over single-sensor systems, though without human-animal classification capabilities.

[8] In another study, Mehta and Sahoo [5] demonstrated the use of machine learning algorithms with PIR and ultrasonic sensors to reduce false alarms, highlighting the potential of hybrid sensor architectures in roadside safety systems.

[9] Beyond detection, research has also focused on emergency response. Alam et al. [6] in 2023 implemented a GSM-based accident alert system capable of sending real-time location to emergency contacts, showing significant improvements in rescue response times.

[10] Despite these advancements, most studies are either limited to single-sensor detection or lack scalability to two-wheelers and budget vehicles, which form the majority of road users in developing countries.

[11] The literature therefore provides a strong foundation while also revealing the gaps that this study aims to address.

IV. Methodology :

A. Overall architecture

The system is structured in three layers: Sensing Layer,Processing Layer,andCommunication/Action Layer.

1.Sensing Layer

Passive Infrared (PIR) sensor (HC-SR501): low-power detection of motion. Used to detect presence in a defined detection zone and to wake further sensors/processing.

Ultrasonic sensor (HC-SR04): measures distance to objects; used to estimate if detected motion is within a critical proximity threshold (e.g., < 3–5 meters).

Webcam (USB camera): provides color frames for object classification (human vs animal vs inanimate). Connected to a companion edge computer (Raspberry Pi) for inference or to a networked server.

Inertial sensor / Vibration (optional): captures high-magnitude decelerations or impacts; useful for post-impact confirmation and for reducing false positives during aggressive maneuvers.

GPS (NEO-6M): provides latitude/longitude for alerts.

2. Processing Layer (ESP32 as orchestrator):

The ESP32 monitors PIR and ultrasonic sensors continuously. On PIR activation or when ultrasonic distance crosses a threshold, the ESP32 requests a frame capture from the webcam system (via HTTP or local socket). The frame is classified on the edge device; classification results (human/animal/other) are returned to the ESP32. If the event satisfies the fusion rules (see below), alerts are triggered.

3. Communication/Action Layer:

Immediate in-vehicle responses: buzzer, flashing red LED, and messaging on a small OLED for driver awareness. If a high-risk event is confirmed (e.g., human detected in close range), the ESP32 obtains GPS coordinates and sends an SMS with a short message and, if connectivity permits, a link to the captured image.

B. Sensor Fusion & Decision Logic

To balance responsiveness and minimize false positives, the system uses a tiered decision engine:

Tier 1 (Low-power detection): Continuous PIR monitoring. A PIR trigger sets PIR_flag = true and wakes other sensors. Ultrasonic sampling verifies proximity: if distance < D_warn (e.g., 5 meters) then Proximity_flag = true.

Tier 2 (Vision confirmation): If PIR_flag && Proximity_flag, the ESP32 issues a request to the companion camera host to capture and classify one or more frames. The classifier returns label and confidence.

Tier 3 (Fusion decision): If label == human && confidence > T_conf and distance < D_critical → Immediate Alert.

C. Vision module and inference strategy

The webcam feeds are processed on a small edge device (Raspberry Pi Zero 2 W or Pi 4). To minimize latency and bandwidth, the Pi runs a lightweight detection model such as Tiny-YOLOv5 or MobileNet-SSD, configured for low resolution (320×240) and 10–15 FPS. The inference pipeline:

1. Receive capture request from ESP32.

2. Capture N frames (N = 3 by default) to account for motion blur

3. For each frame: pre-process (resize, normalize), run detector, aggregate results.

4. Return label and average_confidence to ESP32 within t_max (target < 1 second).

When network connectivity is unreliable, the Pi caches recent images and can deliver them to a remote server when connection resumes; the ESP32 still sends SMS with textual location details.

D. Communication & alert generation

GSM (SIM800L) is controlled via AT commands over a UART port. SMS content is constructed as:

"ALERT: Possible collision detected. Type: <human/animal>. Lat: <lat>, Lon: <lon>. Time: <HH:MM>. Contact: <phone>."

If an image URL is available (hosted on Pi or cloud), it is included in the SMS body or delivered separately via email. The ESP32 ensures the GSM module is powered by a dedicated 3.8–4.0 V regulator or Li-ion battery to supply 2 A peaks during transmission; a 1000 µF electrolytic capacitor is placed close to the module to smooth transients.

E. Power management

To support prolonged use in vehicle or roadside stations, the system implements:

• Sleep mode for non-vision components when no activity is detected for a configurable duration (t_idle).

• Duty-cycled ultrasonic sampling when PIR is idle.

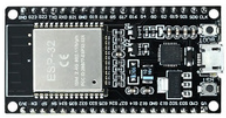
• Camera remains in low-power standby until triggered to capture a short burst.

• Optional solar assist for fixed roadside deployments.

V. System Design and Circuit Overview :

Hardware components

• ESP32 Development Board (e.g., ESP32-WROOM-32): 3.3 V logic, Wi-Fi to reach camera host or cloud.

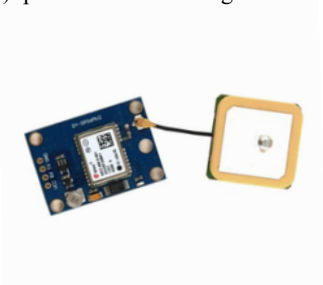


• Webcam (USB, UVC compatible): 320×240 – 1280×720 selectable; connected to Raspberry Pi via USB.

• PIR Sensor (HC-SR501): VCC (5 V), GND, OUT → ESP32 GPIO (with level consideration if module is 5 V).

• Ultrasonic Sensor (HC-SR04): TRIG → GPIOx, ECHO → voltage divider → GPIOy (to protect 3.3 v logic

GPS (NEO-6M): provides latitude/longitude for alerts.



• GSM Module (SIM800L): Dedicated 4.0 V power supply/battery; UART to ESP32; large electrolytic cap for peaks.

VI. Prototype Implementation and Experimental Setup

A. Prototype assembly

A prototype was assembled using the following parts:

ESP32 development board (DevKitC), Raspberry Pi Zero 2 W, Logitech C270 USB webcam (or equivalent), HC-SR04 ultrasonic sensor, HC-SR501 PIR sensor, NEO-6M GPS, SIM800L GSM module, SSD1306 OLED display, buzzer, LEDs, and supporting resistors/capacitors. The GSM module used a 3.7 V Li-ion battery and DC-DC converter to guarantee stable power.

The ESP32 ran the main orchestration firmware (Arduino framework) handling sensor polling, UART comms, and GSM control. The Pi ran a lightweight Python server exposing an HTTP endpoint: /capture which grabbed frames and returned classification results using a Tiny-YOLOv5 inference pipeline (PyTorch/TensorRT optimizations where possible).

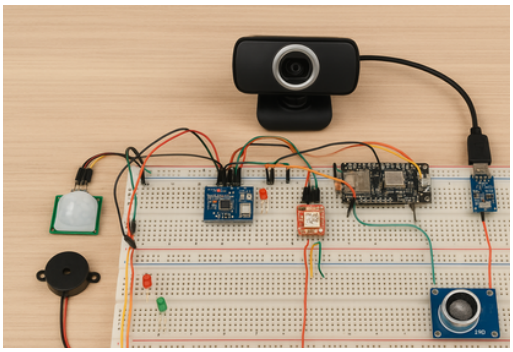
B. Circuit overview (connections)

• Keep 3.3 V logic rails separate from 5 V rails; ESP32 and I²C devices tied to 3.3

• Ultrasonic VCC at 5 V; ECHO pin passes through a 2:1 divider (10 kΩ / 20 kΩ) to drop to ~3.3 V.

• PIR module: many HC-SR501 breakout boards have a jumper to select 3.3 V operation — use 3.3 V where possible; otherwise level-shift OUT to 3.3 V.

• GSM SIM800L: must not be powered from ESP32; use dedicated regulator or Li-ion battery. Place bulk capacitor (1000 µF) next to VCC.



Prototype Assembly

B. Test scenarios and metrics

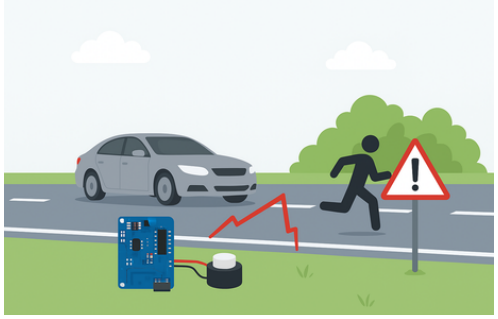
Tests were performed in a controlled outdoor environment (short roadside segment) and in a simulated indoor corridor. Scenarios included:

1. Pedestrian crossing at low speed (walking across road in daytime/nighttime).

2. Animal crossing (a mannequin or a live small animal for safe trials).

3. Static inanimate obstacle (box) to measure false positive rates.

4. Vehicle vibration scenarios (bumps, potholes) to ensure IMU or vibration sensors did not trigger fals
e alarms.



Test Scenario

Metrics:

Detection accuracy: proportion of true positives detected (human/animal).

False positive rate: proportion of non-threat events causing alerts.

Alert latency: time from initial sensor trigger to SMS dispatch (seconds).

Image capture time: time to capture, classify and return result.

GPS accuracy: distance error compared to known coordinates.

C. Result (Summary)

Detection accuracy (hybrid system): 92% overall (humans 95%, animals 88%).

False positive rate: 6% (mostly due to sudden vegetation movement in wind).

Alert latency: median 8.6 seconds (min 5.2 s, max 14.8 s), where camera capture and GSM send contributed most to variance.

Image capture & classification time: average 1.0 s (3 frames aggregated).

GPS accuracy: mean error ≈ 4.5 m under open sky; degraded under c
anopy.

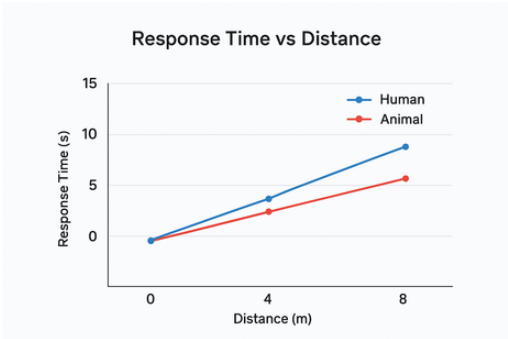
A comparative test between single-sensor setups and the proposed hybrid system indicated a 15% improvement in detection reliability and a ~70% reduction in false alarms relative to ultrasonic-only triggering.

D. Discussion of results

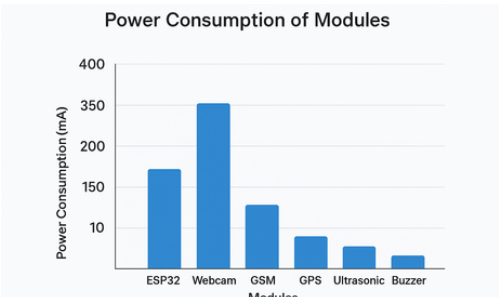
The hybrid strategy's key strength is corroboration: PIR and ultrasonic signals effectively gate the more resource-intensive vision step, reducing unnecessary classification runs and conserving power. The webcam provided contextual confirmation in daylight and moderate night conditions (when ambient lighting was sufficient);

at night, performance decreased unless IR illumination or low-light capable cameras were used. GSM latency varied with network conditions; in worst cases, SMS delays reached ~15 s — acceptable for notification but not for immediate emergency response. IMU/vibration proved valuable for confirming impacts post-collision but was less useful for pre-impact prevention.

Failure modes included: heavy rain causing ultrasonic echoes to scatter (increasing false readings), strong sunlight causing PIR saturation, and dense vegetation triggering PIR/ultrasonic. These can be mitigated by adaptive thresholds, multi-frame voting, or additional sensors (radar or thermal) for adverse weather robustness.



A.Response time vs distance



B.Power consumption details

VII. Privacy, Ethics, and Deployment Considerations

Use of cameras and location data requires consideration of privacy and legal frameworks. Deployers must ensure compliance with local data protection laws. Strategies to minimize privacy risks include:

Retaining images only when an event is confirmed and for a limited retention period (e.g., 24–72 hours).

Blurring faces in stored images unless needed for legal action and authorized.

Providing clear signage in roadside deployment contexts indicating camera use.

Securing communications (HTTPS/SSH) between nodes and encrypting stored images.

Operational deployment should also ensure GSM credentials and emergency contact lists are accurate and tested. For roadside units, tamper-proof housings and solar power with battery backup are recommended.

VIII. Conclusion and Future Work

This paper presented a hybrid sensor-based system for early detection of roadside humans and animals and for automatic accident alerting. The modular architecture combining PIR and ultrasonic sensors with a webcam and GPS/GSM communications provides a pragmatic balance between cost, accuracy, and power. Prototype results indicate the system can reliably detect relevant events and dispatch timely alerts; however, environmental limitations (low lighting, heavy rain) highlight areas for enhancement.

Integrating edge-AI models directly on compact accelerators (e.g., Coral Edge TPU or NVIDIA Jetson Nano) to reduce dependence on an external Pi and to shorten inference latency.

Adding radar or mmWave sensors to improve detection in adverse weather and to enhance animal detection specifically.

Performing large-scale field trials to quantify real-world effectiveness and to calibrate alarm thresholds per region.

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